AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM311
Information Security

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 9/Apr./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **60 marks**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **Not** allowed.

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PART 1: Multiple Choice Questions [10 Marks]

Questions 1 [10 marks - 1 mark/each)]:

1. **Which type of cryptography uses the same key for both encryption and decryption?**
   A) Asymmetric cryptography
   B) Symmetric cryptography
   C) Hashing
   D) Digital signatures
2. **Which of the following is a primary asset in privacy risk assessment?**
   A) Intellectual property
   B) Physical hardware
   C) PII of employees, customers, patients
   D) Software applications
3. **Which category of information processing includes improper protection and handling of PII?**
   A) Aggregation
   B) Exclusion
   C) Identification
   D) Insecurity
4. **Which of the following is an example of personally identifiable information (PII)?**
   A) Birthdate
   B) IP address
   C) Social Security number
   D) All of the above
5. **What is the first stage of privacy by design?**
   A) Privacy planning and policy development
   B) System integration
   C) Implementation and evaluation
   D) Privacy auditing
6. **Which technological advancement has the overall impact of compromising privacy?**
   A) Cloud computing
   B) Decreasing data storage costs
   C) Internet of Things (IoT)
   D) All of the above
7. **Which type of access control ensures that users have the minimum level of access necessary to perform their job functions?**
   A) Role-based access control
   B) Mandatory access control
   C) Discretionary access control
   D) Least privilege access control
8. **What mechanism can prevent an attacker from altering data in transit?**
   A) Digital certificates
   B) Firewalls
   C) Encryption
   D) Antivirus software
9. **What is the primary risk of indefinite data retention?**
   A) Decreased storage costs
   B) Improved data accuracy
   C) Privacy breaches
   D) Enhanced data security
10. **Which is not a component of NIST's privacy framework?**
    A) Identify
    B) Protect
    C) Destroy
    D) Respond

PART 2: Short Answer Questions [20 Marks]

P2-Q1 [5 marks]: What is meant by "incentivized disclosure" in the context of PII?

P2-Q2 [5 marks]: Explain how cloud infrastructure vulnerabilities can lead to data breaches.

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P2-Q3 [5 marks]: What is the significance of encryption in protecting PII?

P2-Q4 [5 marks]: How can cloud infrastructure vulnerabilities impact data privacy?

PART 3: Problem Solving / Critical Thinking Questions [30 Marks]

P3-Q1 [12 marks]: Given the following access control matrix:

	File_1	File_2	File_3
Ali	Read Write	Write	
Ola		Own Read Write	Write Read
Dan	Read Write		Own Write

P3-Q1-1: draw the access control lists for files 1, 2, and 3.

P3-Q1-2: draw the capability lists for files 1, 2 and 3.

P3-Q2 [6 marks]: Your company collects customer feedback through an online portal. How would you handle the collected data to ensure it respects customers' privacy and complies with global privacy regulations? Justify your answer.

P3-Q3 [6 marks]: What steps would you take if a data breach occurs, exposing sensitive customer information? Justify your answer.

P3-Q4 [6 marks]: Consider a scenario where a company is implementing an Attribute-Based Access Control (ABAC) system to manage access to its cloud-based resources. Model an ABAC policy for a cloud storage system where access to files is determined by the following attributes:

- *Subject Attributes*: Role, Department, and Security Clearance Level.
- *Object Attributes*: File Sensitivity, File Owner, and File Type.
- *Environment Attributes*: Time of Access, Location, and Device Type.

Provide a policy rule that allows access to a file based on these attributes.

End of questions

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